Referee report of *'Fast and flexible inference for spatial extremes'*

Overall

I found the paper interesting, and certainly useful if all the proofs and technicalities are correct. I do have some comments about the methodology that I feel should be addressed.

Comments

- I worry that the title is too general and may miss the specificities of the paper, but I leave this up to the AEs to decide. In particular, the restriction of the new models developed to the asymptotic dependent case does rule out a substantial fraction of spatial extremes.
- Pg 2, l51: 'do not provide any statistical guarantees'. Rödder et al. (2025) seem to suggest there is some justification for these neural Bayes estimators.
- Pg 2. l7: 'Vecchia approximation'. More related to your paper on r-Pareto processes and non-stationarity, Koh et al. (2025) fit an r-Pareto model using the gradient score and the Vecchia approximation in an ML algorithm.
- Pg 5. l44: 'The fact that κ can put mass on $\partial\Omega$ hinders the convergence of X, introducing large bias in estimation'. I find this interesting, and I am not currently aware of an obvious reference for this. In any case, I think it would be nice to show a simulation study showing this bias.
- Some typos I spotted (there are others, I'm sure): pg 18, l41: 'the method'. pg 24, l52: 'is is'
- Section 4.3: I found your notation of $\boldsymbol{\eta}$ a little confusing, because it is not always clear if it is a vector of a scalar. For example in pg. 14 l52: $\eta = \mathbf{0}$, but above in l31: $\boldsymbol{\xi}$ has a one to one relationship with $\boldsymbol{\eta}$ but then $\boldsymbol{\xi} = 0$, which is a scalar?

- Section 5.2: How robust is your estimation procedure to not knowing the true centre of the kernels? In practice I can imagine that knowing the true centres is difficult. Is the $\boldsymbol{\eta}$ parameter vector difficult to estimate?
- Pg 19. l27: What is the intuition for choosing this value of $u = 30 \times D$? I understand this should be large but this feels rather arbitrary.
- Fig 3: Isn't the bias you see from the spectral likelihood concerning? Also shouldn't the existing theory say that the pairwise likelihood approach would give unbiased estimates for all the parameters (including b_1 and b_2)?
- Pg 22. l44: 'There is no bias (or approximation) for r-Pareto processes'. Isn't there the approximation when you use the likelihood of the r-Pareto process with L_1 norm as risk functional to make inference for a different risk functional?
- I am generally a little confused with how exatly you are using this trick of choosing $u > D^{1-1/p}$ to go from the L_1 to L_p. This should be spelled out a bit more. Also what does this mean exactly in practice? Do you have to choose a higher u for higher p (L_p) risk functionals? For the data application, page 24, l38: you choose the 99.95% for both risk functionals L_1 and L_∞ as thats where the estimate shape parameter becomes stable, but how is this linked to the trick of looking at the L_1 pareto process? This is also related to pg 30 l3: 'can be bypassed by simply considering the likelihood of a r pareto process with L_1 risk functional.'
- Relatedly, what threshold did you choose in Fig 3 for the L_3 r Pareto simulation study?
- Pg 27. l12: You chose $u = 10 \times D$ here. What exactly is the guidelines for choosing u more generally? Both in the r Pareto and Max-stable cases? Also, here you have 9 observations! And if I'm not mistaken, for up to 9 parameters? How is this still feasible?
- Is there existing theory on using the spectral likehood's asymptotic variance instead of using the jackknife for uncertainty quantification? This could be nice so you can reduce computation for the uncertainty quantification. But I guess you are quite far from asymptotics if you only use as little as 9 observations.

References

Koh, J., D. Steinfeld, and O. Martius (2025). Using spatial extreme-value theory with machine learning to model and understand spatially compounding weather extremes.

Rödder, A., M. Hentschel, and S. Engelke (2025). Theoretical guarantees for neural estimators in parametric statistics.